HIGHER SECONDARY SECOND YEAR ANSWER KEY

COMPUTER SCIENCE

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1 MARKS

Q. NO	ANSWER	MARKS
1	c) Definition	1
2	a) constructor	1
3	b) LEGB	1
4	c) Half-interval search	1
5	b) Ternary	1
6	a) for	1
7	d) return	1
8	a) Positive or negative numbers	1
9	a) eceicS retupmoC	1
10	d) . (dot)	1
11	b) Chen	1
12	b) SELECT	1
13	a) Modification	1
14	b) OS module	1
15	b) Distinct	1

2 MARKS

Q. NO	ANSWER	MARKS
16	**Pair:** ➢ Pair is a compound structure which is made up of list or Tuple. ➢ Any way of bundling two values together into one can be considered as a pair. **Example:** lst := [10, 20]	1 1
17	The process of arranging the list items in ascending or descending order is called sorting. **Example:** Bubble Sort, Selection Sort, Insertion Sort.	1 1
18	**Output:** 5 10 15	2
19	➢ Slice is a substring of a main string. ➢ A substring can be taken from the original string by using [ ] operator and index or subscript values. ➢ Thus, [ ] is also known as slicing operator. ➢ Using slice operator, you have to slice one or more substrings from a main string. **General format of slice operation: str [start: end]**	2
20	Destructor is also a special method gets executed automatically when an object exit from the scope. ➢ In python, _del_ () method is used as destructor. ➢ It is just opposite to constructor. **General format : def __del__(self):** **<statements>**	2
21	**Command** / **Description** / **Component** Insert / Insert data into table / DML	1

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	Create	To create a tables in the Database	DDL	1

22	**Syntax: cd: < absolute path>** ✓ "cd" command used to change directory and absolute path refers to the complete path where python is installed. **Example : "cd c:\program files\openoffice4\Program"**	1 1
23	Matplotlib can be installed using pip software. Pip is a management software for installing python packages. **Importing Matplotlib using the command:** import matplotlib.pyplot as plt Matplotlib can be imported in the workspace.	1 1
24	**Output:** **False** **welcome**	 1 1

3 MARKS

Q. NO	ANSWER	MARKS

25	**Impure Function:**
	The variables used inside the function may cause side effects though the functions which are not passed with any arguments.
	In such cases the function is called impure function.
	Ex:
	let randomnumber :=
	a := random()
	if a>10 then
	return: a
	else
	return: 10
		2
		
		1

26	**Global Scope:** A variable which is declared outside of all the functions in a program is known as global variable. Global variable can be accessed inside or outside of all the functions in a program. **Example:**	2 1

	Entire Program	Output of the Program
1. a:=10 2. Disp(): 3. a:=7 4. print a 5. Disp() 6. print a	a:=10 Disp(): a:=7 print a Disp(): print a	7 10

On execution of the above code the variable **a** which is defined inside the function displays the value 7 for the function call Disp () and then it displays 10, because **a** is defined in global scope.

27	**Assignment Operator:** In Python, = is a simple assignment operator to assign values to variable.	3

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	❁ There are various compound operators in Python like +=, -=, *=, /=, %=, **= and //= are also available.	
28	**ceil()** / Returns the smallest integer greater than or equal to x. / **Ex:**print(math.ceil(23.6)) Output : 24 **floor()** / Returns the largest integer less than or equal to x. / **Ex:**print(math.floor(23.6)) Output:23	3
29	**Syntax:** remove() → List.remove(element) pop() → List.pop(index of an element) clear() → List.clear()	1 1 1
30	**CARTESIAN PRODUCT (Symbol: X)** ✓ Cross product is a way of combining two relations. The resulting relation contains, both relations being combined. ✓ A X B means A times B, where the relation A and B have different attributes. **Ex:** **A=3, B=2 means AXB=3X2=6.**	2 1
31	**reader( ) method** / The reader method is designed to take each line of the file and make a list of all columns / Using this method one can read data from csv files of different format like quotes (' '),(pip (|), and comma(,). / csv.reader works with list\tuple. / **Syntax:** csv.reader(fileobject,delimiter ,fmtparams) **DictReader class** / DictReader works by reading the first line of the CSV and using each comma separated value in this line as a dictionary key. / DictReader is a class of csv module is used to read a CSV file into a dictionary. / It creates an object which maps data to a dictionary. / **Syntax:** csv.DictReader work with dictionary	3
32	**fetchone()** / The fetch one () method returns the next row of a query result set or none in case there is no row left. / Using while loop and fetchone() method we can display all the records from a table. **fetchmany()** / The fetchmany() method returns the next number of rows (n) of the result set. / Displaying specified number of records is done by using fetchmany().	3
33	**Output:** [1,4,9,16,25]	3

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5 MARKS

Q. NO	ANSWER	MARKS
34 A)	**Pure Function:** ➢ Pure functions are functions which will give exact result when the same arguments are passed.	

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- Pure function does not cause any side effects to its output.
- The return value of the pure functions solely depends on its arguments passed.
- They do not modify the arguments which are passed to them.
- If we call pure functions with same set of arguments, we will always get the same result.

Example:

```
let square x:=
    return: x*x
```

Impure Function:

- Impure functions never assure you that the function will behave the same every time it's called.
- Impure function causes side effects to its output.
- The return value of the impure functions does not solely depend on its arguments passed.
- They may modify the arguments which are passed.
- If we call impure functions with same set of arguments, we might get the different return values.

Example:

```
let randomnumber:=
  a:=random( )
  if a>10 then
      return: a
  else
      return: 10
```

5

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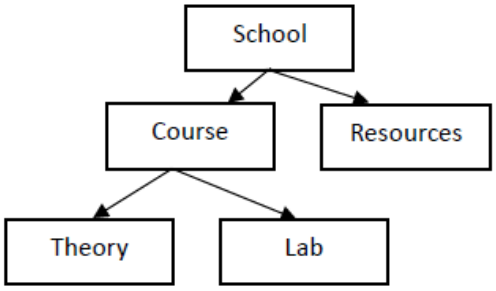
34 B)

Hierarchical Model:

- Hierarchical model was developed by IBM.
- In Hierarchical model, data is represented as a simple tree like structure form. This model represents a one-to-many relationship. i.e., parent-child relationship.
- One child can have only one parent but one parent can have many children. This model is mainly used in IBM Main Frame.



2. Relational Model:

- The Relational Database model was first proposed by E.F.Codd in 1970.
- Nowadays, it is the most widespread data model used for database applications around the world.
- The basic structure of data in relational model is tables (relations).

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- All the information's related to a particular type is stored in rows of that table.
- A relation key is an attribute which uniquely identifies a particular tuple.

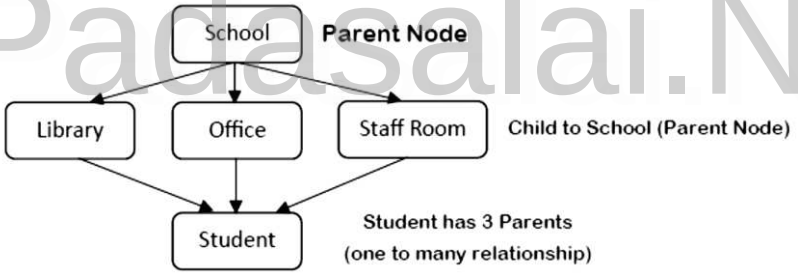
Stu_id	Name	Age
1	Aruna	17
2	Prabhakar	16
3	Aakash	15

Sub_id	Subject	Teacher
1	Tamil	Jegan
2	Maths	Veni
3	Science	Jeya

Stu_id	Sub_id	Marks
1	1	98
2	2	87
3	2	75

3. Network Model:

- Network database model is an extended form of hierarchical data model.
- The difference between hierarchical and Network data model is: In hierarchical model, a child record has only one parent node.
- In a Network model, a child may have many parent nodes. It represents the data in many-to-many relationships.
- This model is easier and faster to access the data.



4. Entity Relationship Model (ER model):

- In this database model, relationships are created by dividing the object into entity and its characteristics into attributes.
- It was developed by Chen in 1976.
- This model is useful in developing a conceptual design for the database.
- It is very simple and easy to design logical view of data.
- The different shapes used in ER diagram is
- Rectangle represents the entities.
  Example: Doctor and Patient
- Ellipse represents the attributes.
  Example: D-id, D-Name, P-id and P-Name
- Diamond represents the relationships.
  Example: Doctor diagnosis the Patient.

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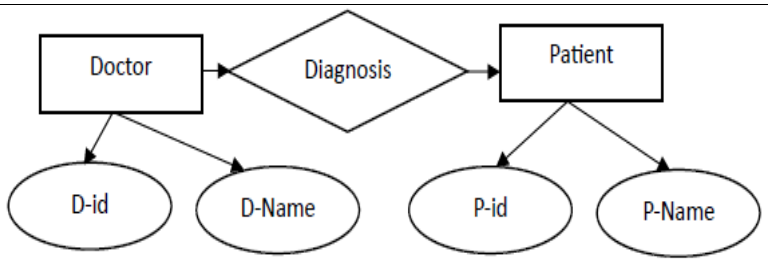
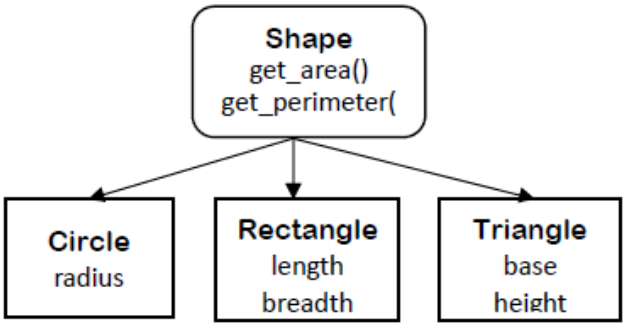


5. Object Model:

- Object model stores the data in the form of objects, attributes and methods, classes and Inheritance.
- This model handles more complex applications, such as Geographic information System (GIS), scientific experiments, engineering design and manufacturing.
- It is used in file Management System.
- It represents real world objects, attributes and behaviours.

Example:



where,

- Shape is the **class**.
- Circle, Rectangle and Triangle are **objects** of class Shape.
- radius, length, breadth, base and height are **attributes**.
- get_area() and get_perimeter() are the **methods**.

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35 A) **Operators in Python:**

In computer programming languages operators are special symbols which represent computations, conditional matching etc.

i. The value of an operator used is called **operands**.
ii. Operators are categorized as Arithmetic, Relational, Logical, Assignment, Conditional etc.
iii. Value and variables when used with operator are known as **operands**. **5**

Arithmetic operators:

1. An arithmetic operator is a mathematical operator that takes two operands and performs a calculation on them.
2. They are used for simple arithmetic.

Relational or Comparative operators:

1. A Relational operator is also called as Comparative operator which checks the relationship between two operands.
2. If the relation is true, it returns True; otherwise it returns False.

Logical operators :

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a. In python, Logical operators are used to perform logical operations on the given relational expressions.
b. There are three logical operators they are and, or and not.

Assignment operators:

a. In Python, = is a simple assignment operator to assign values to variable.
b. There are various compound operators in Python like **+=, -=, *=, /=,** %=, ****=** and **//=** are also available.

Conditional operator or Ternary operator :

- Ternary operator is also known as conditional operator that evaluate something based on a condition being true or false.
- **Syntax**: Variable Name = [on_true] if [Test expression] else [on_false]

35 B)

Constraint:

Constraint is a condition applicable on a field or set of fields

Types of Constraint:

1. Unique Constraint 2.Primary Key Constraint 3.Default Constraint 4. Check Constraint 5. Table constraint

1.Unique Constraint:

- This constraint ensures that no two rows have the same value in the specified columns.
- For example **UNIQUE** constraint applied on Admno of student table ensures that no two students have the same admission number and the constraint can be used as:

CREATE TABLE Student (
Admno integer NOT NULL UNIQUE, → Unique constraint
Name char (20) NOT NULL, Gender char (1),Age integer,
Place char (10));

- The **UNIQUE** constraint can be applied only to fields that have also been declared as **NOT NULL**.
- When two constraints are applied on a single field, it is known as multiple constraints.
- In the above Multiple constraints **NOT NULL** and **UNIQUE** are applied on a single field Admin no.

5

2.Primary Key Constraint:

- This constraint declares a field as a Primary key which helps to uniquely identify a record.
- It is similar to unique constraint except that only one field of a table can be set as primary key.
- The primary key does not allow **NULL** values and therefore a field declared as primary key must have the **NOT NULL** constraint.

Example:

CREATE TABLE Student (
Admno integer NOT NULL PRIMARY KEY, → **Primary Key constraint** Name char(20)NOT NULL, Gender char(1), Age integer, Place char(10));

3.DEFAULT Constraint :

- The **DEFAULT** constraint is used to assign a default value for the field.

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- When no value is given for the specified field having **DEFAULT** constraint, automatically the default value will be assigned to the field.

Example:
CREATE TABLE Student (
Admno integer NOT NULL PRIMARY KEY, Name char(20)NOT NULL, Gender char(1), Age integer DEFAULT = "17", → Default Constraint Place char(10));

- In the above example the "Age" field is assigned a default value of 17, therefore when no value is entered in age by the user, it automatically assigns 17 to Age.

4. Check Constraint :

- This constraint helps to set a limit value placed for a field.
- When we define a check constraint on a single column, it allows only the restricted values on that field.

Example:
CREATE TABLE Student (
Admno integer NOT NULL PRIMARY KEY Name char(20)NOT NULL, Gender char(1), Age integer (CHECK<=19), → Check Constraint Place char(10));

- In the above example the check constraint is set to Age field where the value of age must be less than or equal to 19.

5.TABLE CONSTRAINT :

- When the constraint is applied to a group of fields of the table, it is known as Table constraint.
- The table constraint is normally given at the end of the table definition.
- Let us take a new table namely Student1 with the following fields Admno, First name, Last name, Gender, Age, Place:

CREATE TABLE Student 1 (
Admno integer NOT NULL, Firstname char(20), Lastname char(20), Gender char(1), Age integer, Place char(10),
PRIMARY KEY (Firstname, Lastname) → Table constraint);

- In the above example, the two fields, First name and Last name are defined as Primary key which is a Table constraint.

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36 A)

Syntax:
while <condition>:
statements block 1
[else:
statements block2]

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- ✓ In the **while** loop, the condition is any valid Boolean expression returning True or False.
- ✓ The **else** part of while is optional part of **while**.
- ✓ The **statements block1** is kept executed till the condition is true.
- ✓ If the **else** part is written, it is executed when the condition is tested False.

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	✓ Recall **while** loop belongs to entry check loop type that is it is not executed even once if the condition is tested false in the beginning. **Example:** **Program to print numbers from 10 to 15 using while loop.** i=10 while (i<=15): print (i,end='\t' i=i+1 **Output:** 10 11 12 13 14 15	
36 B)	**Python File Modes:** 1) **'r'** → Open a file for reading (default). 2) **'w'** → Open a file for writing. Creates new file if it does not exist or truncates the file if it exists. 3) **'x'** → Open a file for exclusive creation. If the file already exists, the operation fails. 4) **'t'** → Open in text mode (default). 5) **'a'** → Open for appending at the end of the file without truncating it. Creates a new file if it does not exist. 6) **'b'** → Open in binary mode. 7) **'+'** → Open a file for updating (reading and writing).	**5**
37 A)	**Recursion Function:** • When a function calls itself is known as recursion. • Recursion works like loop but sometimes it makes more sense to use recursion than loop. • You can convert any loop to recursion. A recursive function calls itself. • Imagine a process would iterate indefinitely if not stopped by some condition! Such a process is known as infinite iteration. • The condition that is applied in any recursive function is known as base condition. • A base condition is must in every recursive function otherwise it will continue to execute like an infinite loop. **Overview of how recursive function works:** ❖ Recursive function is called by some external code. ❖ If the base condition is met then the program gives meaningful output and exits. ❖ Otherwise, function does some required processing and then calls itself to continue recursion. ❖ Here is an example of recursive function used to calculate factorial. **Example:** **Output:** def fact(n): 1 if n == 0: 120 return 1 else: return n * fact (n-1) print (fact (0)) T.THIRUMALAI, M.SC(CS).,B.ED., Cell: 9750827717, 7010154722 thirumalaibca.46@gmail.com	**5**
37 B)	**Python's sys module:**	

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- This module provides access to some variables used by the interpreter and to functions that interact strongly with the interpreter.

sys. argv:

- sys.argv is the list of command-line arguments passed to the Python program.
- **argv contains** all the items that come via the command-line input, it's basically a list holding the command line arguments of the program.
- To use **sys.argv**, **import sys** should be used. The first argument, sys. argv [0] contains the name of the python program
- (example pali.py) and sys. argv [1] is the next argument passed to the program (here it is the C++ file).

Python's OS Module :

- The OS module in Python provides a way of using operating system dependent functionality. 5
- The functions that the OS module allows you to interface with the Windows operating system where Python is running on.
- **os.system():** Execute the C++ compiling command (a string contains Unix, C command which also supports C++ command) in the shell (Here it is Command Window).
- For Example to compile C++ program **g++ compiler** should be invoked.

Command:

os.system('g++' + <variable_name1> + '-<mode>' + <variable_name2>)

where each argument contains,

os.system: function system() defined in **os** module to interact with the operating system

g++: General compiler to compile C++ program under windows operating system.

variable_name1: Name of the C++ file along with its path and without extension .cpp in string format

mode: To specify input or output mode. Here it is o prefixed with Hyphen.

variable_name2: Name of the executable file without extension .exe in string format

Python getopt module :

- The getopt module of Python helps you to parse (split) command-line options and arguments.
- This module provides two functions to enable command-line argument parsing.
- This method parses command-line options and parameter list.

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Syntax:

<opts>,<args>=getopt.getopt(argv, options, [long_options])

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Here is the details of the parameters:

1.argv :

→ This is the argument list of values to be parsed (splited). In our program the complete command will be passed as a list.

2.options :

→ This is string of option letters that the Python program recognize as, for input or for output, with options (like 'i' or 'o') that followed by a colon (:). Here colon is used to denote the mode.

3.long_options :

→ This parameter is passed with a list of strings. Argument of Long options should be followed by an equal sign ('=').

getopt() method returns value consisting of two elements. Each of these values are stored separately in two different list (arrays) **opts** and **args**. **args** will be an empty array if there is no error in splitting strings by getopt().

For example:

opts, args = getopt.getopt (argv, "i:",['ifile=''])

where **opts** contains [('-i', 'c:\\pyprg\\p4')]

-i :- option nothing but mode should be followed by : (colon) 'c:\\pyprg\\p4' value nothing but the absolute path of C++ file.

→ In our examples since the entire command line commands are parsed and no leftover argument, the second argument args will be empty [ ].

→ If args is displayed using print () command it displays the output as [ ].

```
>>>print(args)
[ ]
```

38 A)

{1,2,3,4,5,6,7,8}
{4,5}
{1,2,3}
{6,7,8}
{1,2,3,6,7,8}

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38 B)

Aggregate functions of SQL.

- These functions are used to do operations from the values of the column and a single value is returned.

1. COUNT() **2. AVG()** **3. SUM()**
4. MAX() **5. MIN()**

1. COUNT()

- The SQL COUNT() function returns the number of rows in a table satisfying the criteria specified in the WHERE clause.
- COUNT() returns 0 if there were no matching rows.

Example:

```
import sqlite3
connection = sqlite3.connect("Academy.db")
cursor = connection.cursor()
```


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```
cursor.execute("SELECT COUNT(*) FROM student ")
result = cursor.fetchall()
print(result)
```

Output: [(7,)]

2. AVG()

- The following SQL statement in the python program finds the average mark of all students.

Example:

```
import sqlite3
connection = sqlite3.connect("Academy.db")
cursor = connection.cursor()
cursor.execute("SELECT AVG(AVERAGE) FROM student ")
result = cursor.fetchall()
print(result)
```

Output

[(84.65714285714286,)]

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3. SUM():

- The following SQL statement in the python program finds the sum of all average in the Average field of "Student table".

Example:

```
import sqlite3
connection = sqlite3.connect("Academy.db")
cursor = connection.cursor()
cursor.execute("SELECT SUM(AVERAGE) FROM student ")
result = cursor.fetchall()
print(result)
```

Output: [(592.6,)]

4.MAX()

❖ The MAX() function returns the largest value of the selected column.

5.MIN()

❖ The MIN() function returns the smallest value of the selected column.

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